

**18** (a) Show that the equation  $x^3 + x = 7$  has a solution between 1 and 2

(2)

(b) Show that the equation  $x^3 + x = 7$  can be rearranged to give  $x = \sqrt[3]{7 - x}$

(1)

(c) Starting with  $x_0 = 2$ ,  
use the iteration formula  $x_{n+1} = \sqrt[3]{7 - x_n}$  three times to find an estimate for a  
solution of  $x^3 + x = 7$

(3)

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(Total for Question 18 is 6 marks)